

Analytical value of some sixth-order graphs to the electron $g - 2$ in QED

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The contributions of three sixth-order graphs to the electron $g - 2$ in QED, previously known in semi-analytical form, are presented in fully analytical form.

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The QED contribution to the electron anomalous magnetic moment a_e can be expanded as a power series in α/π :

$$a_e^{\text{QED}} = \frac{1}{2} \left[\frac{\alpha}{\pi} \right] + c_2 \left[\frac{\alpha}{\pi} \right]^2 + c_3 \left[\frac{\alpha}{\pi} \right]^3 + c_4 \left[\frac{\alpha}{\pi} \right]^4 + \dots$$

The first two coefficients have been known in (exact) analytical form for a long time; on the other hand, the third and the fourth coefficient are known partially in analytical form and partially in numerical form (actually only a few eight-order graphs contributing to c_4 are known in analytical form).

All known analytical expressions of the contributions to the electron anomaly have a common feature, they are always expressed in simple form (despite the amount of work needed to obtain them) in terms of a few transcendental constants, namely, π^2 , the natural logarithm of 2, the integer values of the Riemann ζ function $\zeta(n) \equiv \sum_{k=1}^{\infty} (1/k^n)$, and the constant $a_4 \equiv \sum_{k=1}^{\infty} (1/2^k k^4)$. This simple behavior suggests that an underlying (not yet discovered) structure may exist.

We bring our attention to the sixth-order graphs shown in Fig. 1, whose contributions are not known in closed analytical form. In fact, Levine, Remiddi, and Roskies [1] worked out the expression of each contribution in “semianalytical” form as a sum of a combination of the above quoted usual transcendental constants plus a number of one-dimensional integrals that they were unable to evaluate analytically (however, the integration was done to high precision by numerical techniques).

In this Brief Report we show how those “residual” integrals can be analytically evaluated and reduced to a combination of the above-quoted transcendental constants. In this way we can also give in closed analytical form the expressions of the contributions of the graphs shown in Fig. 1.

For the convenience of the reader we report here the results from Table I of Ref. [1]. (Due to a misprint, a numerator is missing in the analytical expression of graph 14; the correct expression is taken from Ref. [2]. Infrared pieces proportional to $\ln(\lambda/m)$ and $\ln^2(\lambda/m)$ are omitted, λ and m being, respectively, the infinitesimal photon mass and the electron mass.)

$$a(\text{graph 1}) = \frac{2929}{576} + \frac{431}{432} \pi^2 - \frac{53}{216} \pi^4 + \frac{2}{9} \ln^4 2 + 6\zeta(3) + \frac{16}{3} a_4 - \frac{11}{9} \pi^2 \ln 2 + \frac{16}{9} \pi^2 \ln^2 2 + \frac{689}{96} \zeta(5) - \frac{383}{144} \pi^2 \zeta(3) + \int_0^1 db N_1(b), \quad (1)$$

$$a(\text{graph 4}) = -\frac{7}{48} - \frac{295}{72} \pi^2 - \frac{37}{432} \pi^4 + \frac{1}{18} \ln^4 2 + \frac{629}{12} \zeta(3) + \frac{4}{3} a_4 + \frac{1}{3} \pi^2 \ln 2 + \frac{4}{9} \pi^2 \ln^2 2 - \frac{1153}{96} \zeta(5) - \frac{307}{48} \pi^2 \zeta(3) + \int_0^1 db N_4(b), \quad (2)$$

$$a(\text{graph 14}) = \frac{439}{144} + \frac{311}{144} \pi^2 + \frac{467}{1080} \pi^4 - \frac{4}{9} \ln^4 2 - \frac{1213}{24} \zeta(3) - \frac{32}{3} a_4 - \frac{11}{18} \pi^2 \ln 2 - \frac{32}{9} \pi^2 \ln^2 2 - \frac{347}{96} \zeta(5) + \frac{539}{48} \pi^2 \zeta(3) + \int_0^1 db N_{14}(b), \quad (3)$$

where

$$\begin{aligned} N_1(b) &= I_1(b) + 19I_2(b) + 23I_3(b) - 5I_4(b), \\ N_4(b) &= 23I_1(b) + 29I_2(b) + 121I_3(b) - 523I_4(b), \\ N_{14}(b) &= -23I_1(b) - 65I_2(b) - 157I_3(b) + 487I_4(b), \end{aligned} \quad (4)$$

$$\begin{aligned}
I_1(b) &= \frac{1}{3} \ln(1-b+b^2) [2 \text{Li}_2(b) + 3 \text{Li}_2(-b)] \{ [\ln b - 2 \ln(1-b)] / (1-b) - \ln(1-b)/b \} \\
&\quad + \ln(1-b+b^2) \ln^2(1-b) [2 \ln b - 3 \ln(1+b)] / (3b) \\
&\quad + \ln(1-b+b^2) \ln b \ln(1+b) \ln(1-b)/b + \text{Li}_2(-b) \ln^2(1-b)/b, \\
I_2(b) &= \ln(1-b+b^2) \ln b [4 \text{Li}_2(b) + 6 \text{Li}_2(-b) - 3 \ln(1+b) \ln b] / (12b) - \ln(1-b+b^2) \ln^3(1-b) / (12b), \\
I_3(b) &= \ln(1-b+b^2) \ln^3 b / (6b), \\
I_4(b) &= \ln(1-b+b^2) \ln(1-b) \ln^2 b / (12b)
\end{aligned} \tag{5}$$

and $\text{Li}_2(b)$ is the Euler dilogarithm, defined as

$$\text{Li}_2(b) \equiv - \int_0^b \frac{\ln(1-t)}{t} dt.$$

The difficulty of the analytical integration of functions $I_k(b)$ [and of their linear combinations $N_k(b)$] arises from the presence of the factor $\ln(1-b+b^2)$. In fact, while a complete table of definite integrals of products of polylogarithms up to degree 4 of argument b and $(1 \pm b)$ was already worked out [3] (see Table IV of Ref. [1] for a partial list), the extension to integrals containing the factor $\ln(1-b+b^2)$ is far from complete. Thus, we tried to apply some techniques already developed for the analytical calculation of the contribution of the light-light graphs to the muon $g-2$ [4]. We replace the factor $\ln(1-b+b^2)$ in Eq. (5) with $\ln[1+(x+1/x)b+b^2]$, where x is an additional parameter and call $I_k(b, x)$ the corresponding functions; we then introduce the x -dependent integrals

$$A_k(x) = \int_0^1 db I_k(b, x); \tag{6}$$

the original integrals containing the factor $\ln(1-b+b^2)$ are recovered by setting $x = e^{i(2/3)\pi}$. The explicit analytical expressions of the functions $A_k(x)$ are worked out by repeated differentiation of Eq. (6) with respect to x until the resulting b integrals do not contain polylogarithms and the integration becomes straightforward. Then the $A_k(x)$ are reconstructed from their derivatives by repeated quadratures in x . The constants in the quadratures are fixed at $x=1$ where $\ln[1+(x+1/x)b+b^2]$ becomes $2 \ln(1+b)$ and $A_k(1)$ are known in analytical form [3]. $A_k(x)$ are expressed, at last, as a sum of pentalogarithmic functions.

Once the expressions of the $A_k(x)$ are known, the values of the desired integrals are given by

$$\int_0^1 db I_k(b) = A_k(e^{i(2/3)\pi});$$

the values at $x = e^{i(2/3)\pi}$ of the polylogarithmic functions

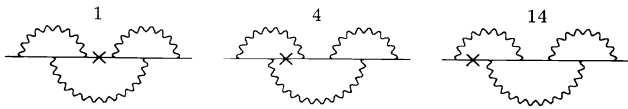


FIG. 1. Graphs evaluated in this paper. The numbering follows Ref. [1].

occurring are worked out using known identities and properties of these functions [5].

The results are

$$\begin{aligned}
\int_0^1 I_1(b) db &= -\frac{4}{27} \pi^3 \sqrt{3} \beta_2 - \frac{24}{17} \pi \sqrt{3} \beta_4 \\
&\quad - \frac{115}{432} \pi^2 \zeta(3) + \frac{49}{3} \zeta(5), \\
\int_0^1 I_2(b) db &= \frac{1}{162} \pi^3 \sqrt{3} \beta_2 + \frac{1}{17} \pi \sqrt{3} \beta_4 \\
&\quad + \frac{10}{81} \pi^2 \zeta(3) - \frac{5281}{2592} \zeta(5), \\
\int_0^1 I_3(b) db &= \frac{25}{27} \zeta(5), \\
\int_0^1 I_4(b) db &= -\frac{1}{162} \pi^3 \sqrt{3} \beta_2 - \frac{1}{17} \pi \sqrt{3} \beta_4 \\
&\quad - \frac{7}{324} \pi^2 \zeta(3) + \frac{287}{324} \zeta(5),
\end{aligned} \tag{7}$$

where

$$\begin{aligned}
\beta_2 &\equiv \frac{\pi^2}{9} - 2 \sum_{k=1}^{\infty} \frac{1}{(6k+1)^2} = 0.976\,628\,016 \dots, \\
\beta_4 &\equiv \frac{5\pi^4}{486} - 2 \sum_{k=1}^{\infty} \frac{1}{(6k+1)^4} = 0.998\,777\,285 \dots
\end{aligned}$$

[we have of course checked that the analytical results of Eq. (7) are in agreement with the numerical evaluations of the corresponding integrals].

From Eqs. (4) and (7) we have for the integrals of Eqs. (1)–(3):

$$\begin{aligned}
\int_0^1 db N_1(b) &= -\frac{529}{96} \zeta(5) + \frac{35}{16} \pi^2 \zeta(3), \\
\int_0^1 db N_4(b) &= -\frac{1109}{32} \zeta(5) + \frac{1261}{144} \pi^2 \zeta(3), \\
\int_0^1 db N_{14}(b) &= \frac{1369}{32} \zeta(5) - \frac{1789}{144} \pi^2 \zeta(3).
\end{aligned} \tag{8}$$

Note that the terms containing the unexpected transcendental constants β_2 and β_4 cancel out from these results.

Using Eq. (8) the contributions of the graphs shown in Fig. [1], Eqs. (1)–(5), are expressed as [we recall that $\zeta(3) = 1.202\,056\,903 \dots$, $\zeta(5) = 1.036\,927\,755 \dots$, $a_4 = 0.517\,479\,061 \dots$]

$$\begin{aligned}
 a(\text{graph } 1) = & \frac{2929}{576} + \frac{431}{432}\pi^2 - \frac{53}{216}\pi^4 + \frac{2}{9}\ln^4 2 + 6\zeta(3) \\
 & + \frac{16}{3}a_4 - \frac{11}{9}\pi^2\ln 2 + \frac{16}{9}\pi^2\ln^2 2 + \frac{5}{3}\zeta(5) \\
 & - \frac{17}{36}\pi^2\zeta(3) = -2.751\,419\,526\ldots, \quad (9)
 \end{aligned}$$

$$\begin{aligned}
 a(\text{graph } 4) = & -\frac{7}{48} - \frac{295}{72}\pi^2 - \frac{37}{432}\pi^4 \\
 & + \frac{1}{18}\ln^4 2 + \frac{629}{12}\zeta(3) \\
 & + \frac{4}{3}a_4 + \frac{1}{3}\pi^2\ln 2 + \frac{4}{9}\pi^2\ln^2 2 - \frac{140}{3}\zeta(5) \\
 & + \frac{85}{36}\pi^2\zeta(3) = -1.206\,376\,517\ldots, \quad (10)
 \end{aligned}$$

$$\begin{aligned}
 a(\text{graph } 14) = & \frac{439}{144} + \frac{311}{144}\pi^2 + \frac{467}{1080}\pi^4 \\
 & - \frac{4}{9}\ln^4 2 - \frac{1213}{24}\zeta(3) \\
 & - \frac{32}{3}a_4 - \frac{11}{8}\pi^2\ln 2 - \frac{32}{9}\pi^2\ln^2 2 + \frac{235}{6}\zeta(5) \\
 & - \frac{43}{36}\pi^2\zeta(3) = 5.509\,933\,641\ldots \quad (11)
 \end{aligned}$$

The total sum is

$$\begin{aligned}
 a(1+4+14) = & \frac{4601}{576} - \frac{203}{216}\pi^2 + \frac{73}{720}\pi^4 \\
 & - \frac{1}{6}\ln^4 2 + \frac{63}{8}\zeta(3) \\
 & - 4a_4 - \frac{3}{2}\pi^2\ln 2 - \frac{4}{3}\pi^2\ln^2 2 - \frac{35}{6}\zeta(5) \\
 & + \frac{25}{36}\pi^2\zeta(3) = 1.552\,137\,597\ldots \quad (12)
 \end{aligned}$$

The numerical values of Eqs. (9)–(12) are in perfect agreement with the corresponding numerical values shown in Table I of Ref. [1].

All the algebraic manipulations were carried out through the symbolic manipulation program ASHMEDAI [6].

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